

A Model-Based Attitude and Heading Reference System Utilizing a Bio-Inspired Polarized Sensor

This repository contains the source code, simulation data, and experimental test files for the paper provided in the reference file [PAHRS_Report.pdf](#). The paper, titled "A Model-Based Attitude and Heading Reference System Utilizing a Bio-Inspired Polarized Sensor", proposes a framework that integrates a bio-inspired polarized skylight sensor to overcome the limitations of conventional attitude and heading reference systems.

The files related to the experimental tests and simulations are categorized into the following four main sections:

1. **Simulation**
2. **Experimental Test on a 3-DOF Quadrotor Stand**
3. **Experimental Test with a PX4-Based Quadrotor Flight**
4. **Optimized Code for GPU Implementation (Real-Time Capability Analysis)**

1. Simulation

This section consists of three sub-modules:

I. Data Generation

Located in the Data Generation folder, the script [Data_Generation_Realtime.ipynb](#) takes the CSV file inside the Simulation_Data directory as its input and outputs the Polarization data results into the Simulation_Results folder.

II. Heading (Psi) Measurement

The [Psi_Measurement.ipynb](#) script takes the data simulation results from the Data Generation step as input, measures the heading angle, and saves the output in the Results folder.

III. Model-Aided Navigation Simulation

To run this simulation, you must first execute the [Initialization.m](#) script to provide the necessary inputs for the Simulink model [ModelAided_PAHRS.slx](#). The heading angle

results from the Heading Measurement section are stored in the [yaw_mes.mat](#) file, which is loaded by [Initialization.m](#). After initialization, run the Simulink file, and then execute [Results.m](#) to display the simulation outputs.

2. Experimental Test on a 3-DOF Quadrotor Stand

The main script for this section is [PAHRS.m](#).

- Input: It loads the [test_data.mat](#) file internally.
- Execution: Run [PAHRS.m](#) first, followed by the [Results.m](#) script to visualize the outputs.
- Note: The MAHRS-related results are already saved in the [MAHRS_Results.mat](#) file.

3. Experimental Test with a PX4-Based Quadrotor Flight

The main script for this section is also [PAHRS.m](#).

- Input: It loads the [log_data.mat](#) file internally.
- Execution: Run [PAHRS.m](#) first, followed by the [Results.m](#) script to visualize the outputs.
- Note: The MAHRS-related results are already saved in the [MAHRS_Results.mat](#) file.

4. Optimized Code for GPU Implementation (Real-Time Capability Analysis)

This section contains the optimized code designed to verify real-time processing capabilities. The optimized script [Psi_Measurement_Optimized.ipynb](#), which takes the folder 2026-06-17_19-34-06 as its input, is configured to be executed directly on a GPU.